Project 1 Report

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Overview

This project focuses on setting up the required environment for conducting computer architecture experiments. Specifically, we use ChampSim to simulate processor microarchitecture and the memory hierarchy. The first step involves setting up a manual environment in WSL2 (Ubuntu 22.04) instead of Docker, and installing ChampSim within it. During the installation process, we clone the ChampSim code repository from GitHub, resolve dependencies using vcpkg, and compile it. Afterward, we configure a customized processor system using a JSON configuration file and execute a specific instruction trace (binary_search) to evaluate the performance of the simulated system.

Configuration and Compilation

We successfully set up a manual environment in WSL2 (Ubuntu 22.04) instead of using Docker. We used the provided basic configuration file `lab1_config.json` and resolved all dependencies via `vcpkg` (which involved manually downloading packages due to GitHub network constraints). We then compiled the source code by running `./config.sh lab1_config.json` and `make -j $(nproc)`. The successful compilation generated the executable `bin/champsim-lab1`. A screenshot of the successful configuration and compilation is shown below.



Figure 1: Successful configuration and compilation of ChampSim.

Simulation and Results

Short Simulation

To verify that the simulator works correctly, we ran a short simulation with the `binary_search` trace. We used 1,000,000 warmup instructions and 1,000,000 simulation instructions. The output was saved using the `tee` command for further analysis. The command used is shown below:

```
./bin/champsim-lab1 --warmup-instructions 1000000 \
                    --simulation-instructions 1000000 \
                    traces/binary_search.champsimtrace.xz \
                    | tee results-part1-short.txt
```

The simulation finished without errors and successfully generated the performance statistics. A screenshot of the successful execution is shown below.

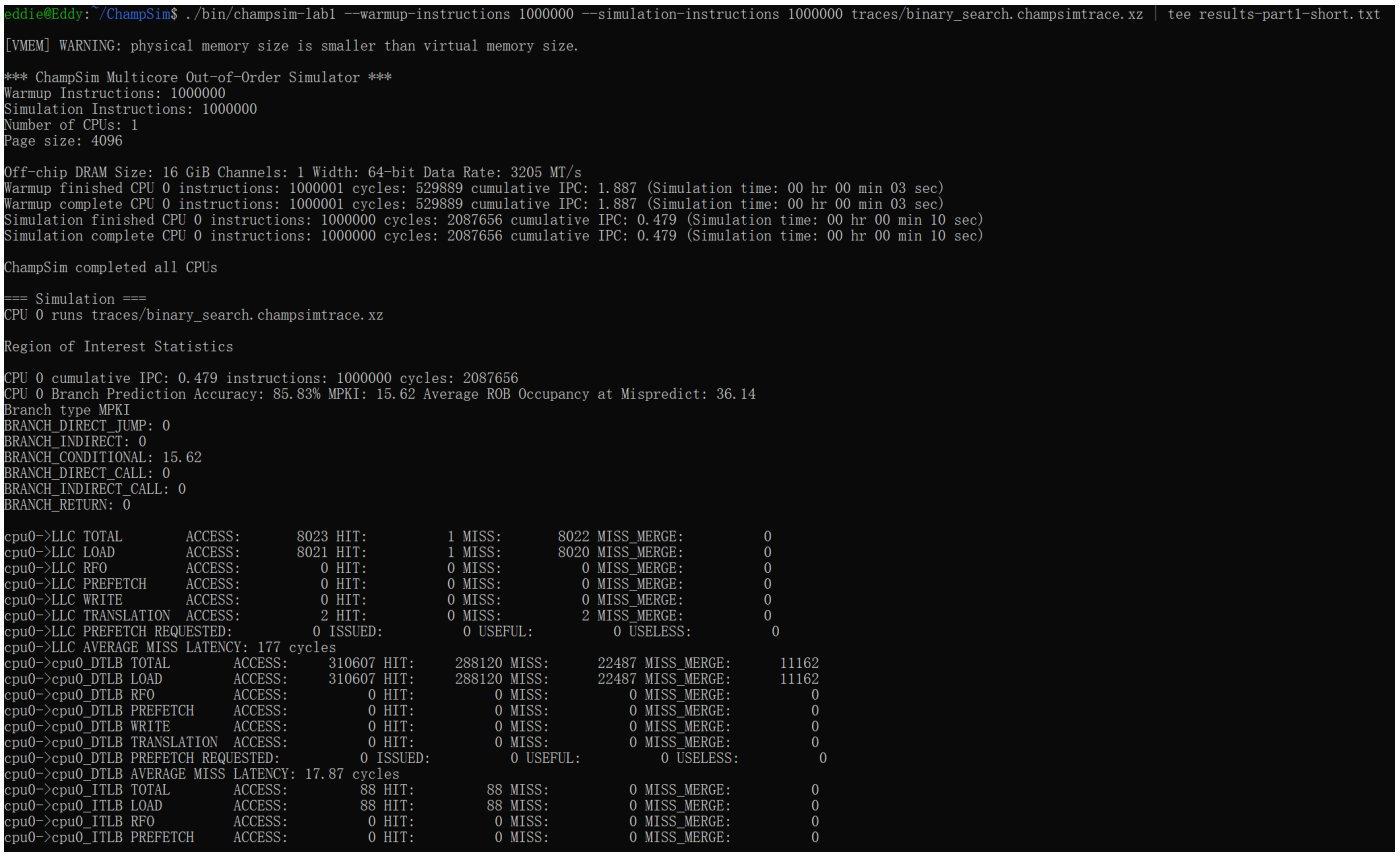


Figure 2: Successful execution of ChampSim with the binary_search trace (Short test).

Long Simulation

After verifying the simulator with the short test, we ran a long simulation to obtain representative performance data for the report. We used 10,000,000 warmup instructions and 50,000,000 simulation instructions. The command used is shown below:

```
./bin/champsim-lab1 --warmup-instructions 10000000 \
                    --simulation-instructions 50000000 \
                    traces/binary_search.champsimtrace.xz \
                    | tee results-part1-short.txt
```

A screenshot of the successful long simulation is shown below.

```
eddie@Eddy:~/ChampSim$ ./bin/champsim-lab1 --warmup-instructions 10000000 --simulation-instructions 50000000 traces/binary_search.champsimtrace.xz | tee results-part1-long.txt
[VMEM] WARNING: physical memory size is smaller than virtual memory size.

*** ChampSim Multicore Out-of-Order Simulator ***
Warmup Instructions: 10000000
Simulation Instructions: 50000000
Number of CPUs: 1
Page size: 4096

Off-chip DRAM Size: 16 GiB Channels: 1 Width: 64-bit Data Rate: 3205 MT/s
Heartbeat CPU 0 instructions: 10000004 cycles: 4125516 heartbeat IPC: 2.424 cumulative IPC: 2.424 (Simulation time: 00 hr 00 min 29 sec)
Warmup finished CPU 0 instructions: 10000004 cycles: 4125516 cumulative IPC: 2.424 (Simulation time: 00 hr 00 min 29 sec)
Warmup complete CPU 0 instructions: 10000004 cycles: 4125516 cumulative IPC: 2.424 (Simulation time: 00 hr 00 min 29 sec)
Heartbeat CPU 0 instructions: 20000004 cycles: 21307595 heartbeat IPC: 0.582 cumulative IPC: 0.582 (Simulation time: 00 hr 01 min 25 sec)
Heartbeat CPU 0 instructions: 30000004 cycles: 38541432 heartbeat IPC: 0.5803 cumulative IPC: 0.5811 (Simulation time: 00 hr 02 min 23 sec)
Heartbeat CPU 0 instructions: 40000005 cycles: 55801749 heartbeat IPC: 0.5794 cumulative IPC: 0.5805 (Simulation time: 00 hr 03 min 21 sec)
Heartbeat CPU 0 instructions: 50000005 cycles: 72991028 heartbeat IPC: 0.5818 cumulative IPC: 0.5808 (Simulation time: 00 hr 04 min 21 sec)
Heartbeat CPU 0 instructions: 60000005 cycles: 90178919 heartbeat IPC: 0.5818 cumulative IPC: 0.581 (Simulation time: 00 hr 05 min 30 sec)
Simulation finished CPU 0 instructions: 50000001 cycles: 86053403 cumulative IPC: 0.581 (Simulation time: 00 hr 05 min 30 sec)
Simulation complete CPU 0 instructions: 50000001 cycles: 86053403 cumulative IPC: 0.581 (Simulation time: 00 hr 05 min 30 sec)

ChampSim completed all CPUs

=== Simulation ===
CPU 0 runs traces/binary_search.champsimtrace.xz

Region of Interest Statistics

CPU 0 cumulative IPC: 0.581 instructions: 50000001 cycles: 86053403
CPU 0 Branch Prediction Accuracy: 85.81% MPKI: 15.65 Average ROB Occupancy at Mispredict: 35.99
Branch type MPKI
BRANCH_DIRECT_JUMP: 0
BRANCH_INDIRECT: 0
BRANCH_CONDITIONAL: 15.65
BRANCH_DIRECT_CALL: 0
BRANCH_INDIRECT_CALL: 0
BRANCH_RETURN: 0

cpu0->LLC TOTAL        ACCESS:     300169 HIT:      17823 MISS:     282346 MISS_MERGE:          0
cpu0->LLC LOAD         ACCESS:     299847 HIT:      17813 MISS:     282034 MISS_MERGE:          0
cpu0->LLC RFO          ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->LLC PREFETCH     ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->LLC WRITE        ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->LLC TRANSLATION  ACCESS:        322 HIT:         10 MISS:        312 MISS_MERGE:          0
cpu0->LLC PREFETCH REQUESTED:          0 ISSUED:          0 USEFUL:          0 USELESS:          0
cpu0->LLC AVERAGE MISS LATENCY: 174.4 cycles
cpu0->cpu0_DTLB TOTAL        ACCESS:   15433837 HIT:   14319968 MISS:    1113869 MISS_MERGE:     547823
cpu0->cpu0_DTLB LOAD         ACCESS:   15433837 HIT:   14319968 MISS:    1113869 MISS_MERGE:     547823
cpu0->cpu0_DTLB RFO          ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_DTLB PREFETCH     ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_DTLB WRITE        ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_DTLB TRANSLATION  ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_DTLB PREFETCH REQUESTED:          0 ISSUED:          0 USEFUL:          0 USELESS:          0
cpu0->cpu0_DTLB AVERAGE MISS LATENCY: 5.473 cycles
cpu0->cpu0_ITLB TOTAL        ACCESS:       5075 HIT:       5075 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_ITLB LOAD         ACCESS:       5075 HIT:       5075 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_ITLB RFO          ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_ITLB PREFETCH     ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_ITLB WRITE        ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_ITLB TRANSLATION  ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_ITLB PREFETCH REQUESTED:          0 ISSUED:          0 USEFUL:          0 USELESS:          0
cpu0->cpu0_ITLB AVERAGE MISS LATENCY: - cycles
cpu0->cpu0_L1D TOTAL        ACCESS:   15504865 HIT:   13985800 MISS:    1519065 MISS_MERGE:     761662
cpu0->cpu0_L1D LOAD         ACCESS:   11928253 HIT:   10437839 MISS:    1490414 MISS_MERGE:     759410
cpu0->cpu0_L1D RFO          ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_L1D PREFETCH     ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_L1D WRITE        ACCESS:    3505584 HIT:    3505584 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_L1D TRANSLATION  ACCESS:      71028 HIT:      42377 MISS:      28651 MISS_MERGE:       2252
cpu0->cpu0_L1D PREFETCH REQUESTED:          0 ISSUED:          0 USEFUL:          0 USELESS:          0
cpu0->cpu0_L1D AVERAGE MISS LATENCY: 83.51 cycles
cpu0->cpu0_L1I TOTAL        ACCESS:       5075 HIT:       5075 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_L1I LOAD         ACCESS:       5075 HIT:       5075 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_L1I RFO          ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_L1I PREFETCH     ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_L1I WRITE        ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_L1I TRANSLATION  ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_L1I PREFETCH REQUESTED:          0 ISSUED:          0 USEFUL:          0 USELESS:          0
cpu0->cpu0_L1I AVERAGE MISS LATENCY: - cycles
cpu0->cpu0_L2C TOTAL        ACCESS:     757403 HIT:     457234 MISS:     300169 MISS_MERGE:          0
cpu0->cpu0_L2C LOAD         ACCESS:     731004 HIT:     431157 MISS:     299847 MISS_MERGE:          0
cpu0->cpu0_L2C RFO          ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_L2C PREFETCH     ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_L2C WRITE        ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_L2C TRANSLATION  ACCESS:      26399 HIT:      26077 MISS:        322 MISS_MERGE:          0
cpu0->cpu0_L2C PREFETCH REQUESTED:          0 ISSUED:          0 USEFUL:          0 USELESS:          0
cpu0->cpu0_L2C AVERAGE MISS LATENCY: 182 cycles
cpu0->cpu0_STLB TOTAL        ACCESS:     566046 HIT:     530532 MISS:      35514 MISS_MERGE:          0
cpu0->cpu0_STLB LOAD         ACCESS:     566046 HIT:     530532 MISS:      35514 MISS_MERGE:          0
cpu0->cpu0_STLB RFO          ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_STLB PREFETCH     ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_STLB WRITE        ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_STLB TRANSLATION  ACCESS:          0 HIT:          0 MISS:          0 MISS_MERGE:          0
cpu0->cpu0_STLB PREFETCH REQUESTED:          0 ISSUED:          0 USEFUL:          0 USELESS:          0
cpu0->cpu0_STLB AVERAGE MISS LATENCY: 22.48 cycles

DRAM Statistics

Channel 0 RQ ROW_BUFFER_HIT:        111
  ROW_BUFFER_MISS:     282235
  AVG DBUS CONGESTED CYCLE: 2.12
Channel 0 WQ ROW_BUFFER_HIT:          0
  ROW_BUFFER_MISS:          0
  FULL:          0
Channel 0 REFRESHES ISSUED:       7171
```

Figure 3: Successful execution of ChampSim with the binary_search trace (Long test).

Comparison of Simulation Results

The key performance metrics from both the short and long simulations are summarized in Table 1.

Table 1: Comparison of Short and Long Simulation Results

Metric	Short Simulation (1M/1M)	Long Simulation (10M/50M)
Cumulative IPC	0.479	0.581
Branch Prediction Accuracy	85.83%	85.81%
Branch MPKI	15.62	15.65
L1D Total Accesses	1,650,485	16,504,856
L1D Total Misses	251,985	1,510,696
L1D Miss Rate	15.27%	9.15%
LLC Total Accesses	8,023	300,169
LLC Total Misses	8,022	282,346
LLC Miss Rate	99.99%	94.06%

As shown in Table 1, the long simulation exhibits a more stable performance state. The cumulative IPC increases from 0.479 to 0.581, and the L1D miss rate drops significantly from 15.27% to 9.15% after the longer warmup phase. However, the LLC miss rate remains extremely high (over 94%) due to the random memory access pattern of the `binary_search` workload, which remains the primary bottleneck preventing a higher IPC.

Workflow Description

The complete workflow of ChampSim consists of four main steps: configuration, compilation, trace execution, and statistics collection.

1. **Configuration**: We define the processor microarchitecture parameters in `lab1_config.json`, including CPU core settings, cache hierarchy (L1I, L1D, L2C sets and ways), and the branch predictor type (`hashed_perceptron`).
2. **Compilation**: We run `./config.sh lab1_config.json` to generate the build configuration, followed by `make -j $(nproc)` to compile the source code. This generates the executable `bin/champsim-lab1`.
3. **Trace Execution**: We run the executable with a provided trace file (e.g., `binary_search.champsimtrace.xz`). We specify the number of warmup instructions and simulation instructions using command-line arguments.

4. **Statistics Collection**: After the simulation completes, ChampSim outputs detailed performance statistics (such as IPC, branch prediction accuracy, and cache hit/miss rates) to the terminal. We use the `tee` command to save this output to a text file for further analysis.